Those questions highlighted in yellow are flagged out in the examiner's report as challenging.

General Comments

The standard of the work this year was very good. There were many excellent, well presented, clear responses. The majority of candidates appeared to have enough space within the answer booklet.

Most candidates attempted all the questions, demonstrating a strong knowledge and understanding of most of the topics. Many candidates performed well on the contextual questions, in particular **Question 11.**

Question 5, relating to vectors, indicated that not all candidates could select and use appropriate methods when solving the problem. Selecting and using appropriate mathematical language was also challenging for some candidates.

Candidates should be encouraged to consider the mark allocation for questions. Some candidates gave lengthy solutions to problems when the marks available indicated that a simpler method was expected.

Candidates did not always provide exact solutions when asked to do so, for example in **Question 7(a)(ii)**.

1 A plane π_1 contains two vectors $\begin{pmatrix}1\\1\\0\end{pmatrix}$ and $\begin{pmatrix}1\\-5\\-2\end{pmatrix}$.

(i) Find a vector normal to π_1. [2]

Solutions	Comments
$\begin{pmatrix}1\\1\\0\end{pmatrix}\times\begin{pmatrix}1\\-5\\-2\end{pmatrix}=\begin{pmatrix}-2\\2\\-6\end{pmatrix}=2\begin{pmatrix}-1\\1\\-3\end{pmatrix}$ A vector normal to π_1 is $\begin{pmatrix}-1\\1\\-3\end{pmatrix}$.	Any vector $k\begin{pmatrix}-1\\1\\-3\end{pmatrix}$, $k\in\mathbb{R}$ is also a normal to π_1. Note that $\begin{pmatrix}-2\\2\\-6\end{pmatrix}\neq\begin{pmatrix}-1\\1\\-3\end{pmatrix}$

A plane π_2 has equation $4x+5y-6z=0$.

(ii) Find the acute angle between π_1 and π_2. [2]

Solutions	Comments
Acute angle between π_1 and π_2 $=\cos^{-1}\dfrac{\begin{pmatrix}-1\\1\\-3\end{pmatrix}\cdot\begin{pmatrix}4\\5\\-6\end{pmatrix}}{\sqrt{1+1+9}\sqrt{16+25+36}}$ $=49.2^\circ$ (to 1 d.p.)	To find the angle between the planes, consider the angle between the normals

2 A curve has equation $\dfrac{x^2}{1+x^2}+\dfrac{y^2}{1+y^2}=x^3y^5$. Find the equation of the tangent to the curve at the point (1, 1). Give your answer in the form $ax+by=c$, where a, b and c are integers. [6]

Solutions	Comments
$\dfrac{x^2}{1+x^2}+\dfrac{y^2}{1+y^2}=x^3y^5$ $1-\dfrac{1}{1+x^2}+1-\dfrac{1}{1+y^2}=x^3y^5$ $2x\left(1+x^2\right)^{-2}+2y\left(1+y^2\right)^{-2}\dfrac{\mathrm{d}y}{\mathrm{d}x}=3x^2y^5+5x^3y^4\dfrac{\mathrm{d}y}{\mathrm{d}x}$ At (1, 1): $\dfrac{1}{2}+\dfrac{1}{2}\dfrac{\mathrm{d}y}{\mathrm{d}x}=3+5\dfrac{\mathrm{d}y}{\mathrm{d}x}$ $\dfrac{\mathrm{d}y}{\mathrm{d}x}=-\dfrac{5}{9}$ Equation of tangent: $y-1=-\dfrac{5}{9}(x-1)$ $5x+9y=14$	Alternatively, you may apply quotient rule to differentiate. You should not attempt to find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ in terms of x and y. Instead, you should immediately substitute $x=1$ and $y=1$ after differentiation without any algebraic manipulation. Remember to give the equation in the correct form, leaving coefficients as integers as required.

Similar to 2018/I/Q7

3 It is given that $\mathrm{f}(x)=\ln(1+\sin 3x)$.

(i) Show that $\mathrm{f}''(x)=\dfrac{k}{1+\sin 3x}$, where k is a constant to be found. [3]

Solutions	Comments
$\mathrm{f}(x)=\ln(1+\sin 3x)$ $\mathrm{f}'(x)=\dfrac{3\cos 3x}{1+\sin 3x}$ $\mathrm{f}''(x)=\dfrac{(1+\sin 3x)(-9\sin 3x)-(3\cos 3x)(3\cos 3x)}{(1+\sin 3x)^2}$ $=\dfrac{-9\sin 3x-9\sin^2 3x-9\cos^2 3x}{(1+\sin 3x)^2}$ $=\dfrac{-9(\sin 3x+\sin^2 3x+\cos^2 3x)}{(1+\sin 3x)^2}$ $=\dfrac{-9(1+\sin 3x)}{(1+\sin 3x)^2}$ $=\dfrac{-9}{1+\sin 3x}$	Remember to apply Chain Rule when differentiating $\sin 3x$ You would need to apply your quotient rule correctly and be careful in your algebraic manipulation.

(ii) Hence find the first three non-zero terms of the Maclaurin expansion of $\mathrm{f}(x)$. [4]

Solutions	Comments
From part **(i)**, $\mathrm{f}(0)=0$, $\mathrm{f}'(0)=3$, $\mathrm{f}''(0)=-9$, and $\mathrm{f}'''(x)=\dfrac{9(3\cos 3x)}{(1+\sin 3x)^2}=\dfrac{27\cos 3x}{(1+\sin 3x)^2}$ gives $\mathrm{f}'''(0)=27$. $\therefore \mathrm{f}(x)=0+3x-\dfrac{9}{2!}x^2+\dfrac{27}{3!}x^3+\ldots$ $=3x-\dfrac{9}{2}x^2+\dfrac{9}{2}x^3+\ldots$	

Similar to 2016/I/Q8 & 2018/II/Q4

4 Do not use a calculator in answering this question.

Three complex numbers are $z_1 = 1+\sqrt{3}\mathrm{i}$, $z_2 = 1-\mathrm{i}$ and $z_3 = 2(\cos\frac{1}{6}\pi + \mathrm{i}\sin\frac{1}{6}\pi)$.

(i) Find $\frac{z_1}{z_2 z_3}$ in the form $r(\cos\theta + \mathrm{i}\sin\theta)$, where $r>0$ and $-\pi<\theta\leqslant\pi$. [4]

Solutions	Comments
$\lvert z_1\rvert = \sqrt{1+3} = 2;\quad \arg(z_1) = \tan^{-1}\frac{\sqrt{3}}{1} = \frac{\pi}{3} \quad \Rightarrow z_1 = 2e^{\frac{\pi}{3}i}$ $\lvert z_2\rvert = \sqrt{1+1} = \sqrt{2};\quad \arg(z_2) = -\tan^{-1}\frac{1}{1} = -\frac{\pi}{4} \quad \Rightarrow z_2 = \sqrt{2}e^{-\frac{\pi}{4}i}$ $z_3 = 2(\cos\frac{1}{6}\pi + \mathrm{i}\sin\frac{1}{6}\pi) = 2e^{\frac{\pi}{6}i}$ $\frac{z_1}{z_2 z_3} = \frac{2e^{\frac{\pi}{3}i}}{\sqrt{2}e^{-\frac{\pi}{4}i}\times 2e^{\frac{\pi}{6}i}}$ $= \frac{1}{\sqrt{2}}e^{\left(\frac{\pi}{3}+\frac{\pi}{4}-\frac{\pi}{6}\right)i}$ $= \frac{1}{\sqrt{2}}e^{\frac{5\pi}{12}i}$ $= \frac{1}{\sqrt{2}}\left(\cos\frac{5\pi}{12} + \mathrm{i}\sin\frac{5\pi}{12}\right)$	Express z_1, z_2 and z_3 in polar exponential form so that it is easier to multiply and divide (to find $\frac{z_1}{z_2 z_3}$). [We usually will use polar exponential form when working with multiplication/division of complex numbers] ensure you check $\arg(z_2)$ is **–ve and acute**, because of the quadrant z_2 is in. Look at how we present the working too, **not** $\tan^{-1}\frac{-1}{1}$, **also not** $\tan^{-1}\frac{1}{1} = -\frac{\pi}{4}$ -- ask your tutor if unsure.

A fourth complex number, z_4, is such that $\frac{z_1 z_4}{z_2 z_3}$ is purely imaginary and $\left|\frac{z_1 z_4}{z_2 z_3}\right| = 1$.

(ii) Find the possible values of z_4 in the form $r(\cos\theta + \mathrm{i}\sin\theta)$, where $r>0$ and $-\pi<\theta\leqslant\pi$. [3]

$\left\lvert\frac{z_1 z_4}{z_2 z_3}\right\rvert = 1 \quad\Rightarrow\quad \left\lvert\frac{z_1}{z_2 z_3}\right\rvert \times \lvert z_4\rvert = 1 \quad\Rightarrow\quad \lvert z_4\rvert = \sqrt{2}$ $\frac{z_1 z_4}{z_2 z_3}$ is purely imaginary $\Rightarrow \quad \arg\left(\frac{z_1 z_4}{z_2 z_3}\right) = \pm\frac{\pi}{2}$ $\arg\left(\frac{z_1}{z_2 z_3}\right) + \arg(z_4) + 2k\pi = \pm\frac{\pi}{2}, \quad k\in\mathbb{Z}$ $\arg(z_4) = \pm\frac{\pi}{2} - \frac{5\pi}{12} + 2k\pi$ For $-\pi<\theta\leqslant\pi$, $k=0$, $\quad \arg(z_4) = \frac{\pi}{12}$ or $-\frac{11\pi}{12}$ $\therefore z_4 = \sqrt{2}\left(\cos\frac{\pi}{12} + \mathrm{i}\sin\frac{\pi}{12}\right)$ or $\sqrt{2}\left(\cos\left(-\frac{11\pi}{12}\right) + \mathrm{i}\sin\left(-\frac{11\pi}{12}\right)\right)$	Recall properties of modulus and argument 1. $\lvert zw\rvert = \lvert z\rvert\lvert w\rvert$ and $\arg(zw) = \arg z + \arg w$ 2. $\left\lvert\frac{z}{w}\right\rvert = \frac{\lvert z\rvert}{\lvert w\rvert}$ and $\arg\left(\frac{z}{w}\right) = \arg z - \arg w$

5 (a) Given that **a** and **b** are non-zero vectors such that $\mathbf{a}\times\mathbf{b}=\mathbf{b}\times\mathbf{a}$, find the relationship between **a** and **b**. [2]

Solutions	Comments
$\underset{\sim}{a}\times\underset{\sim}{b}=\underset{\sim}{b}\times\underset{\sim}{a}$ $\underset{\sim}{a}\times\underset{\sim}{b}-\underset{\sim}{b}\times\underset{\sim}{a}=\underset{\sim}{0}$ $2\left(\underset{\sim}{a}\times\underset{\sim}{b}\right)=\underset{\sim}{0} \qquad \because \underset{\sim}{b}\times\underset{\sim}{a}=-\underset{\sim}{a}\times\underset{\sim}{b}$ $\underset{\sim}{a}\times\underset{\sim}{b}=\underset{\sim}{0}$ Since $\underset{\sim}{a}$ and $\underset{\sim}{b}$ are non-zero, $\underset{\sim}{a}$ is parallel to $\underset{\sim}{b}$.	

(b) The points P, Q and R have position vectors **p**, **q** and **r** respectively. The points P and Q are fixed and R varies.

(i) Given that **q** is non-zero and $(\mathbf{r}-\mathbf{p})\times\mathbf{q}=\mathbf{0}$, describe geometrically the set of all possible positions of the point R. [3]

Solutions	Comments
$\left(\underset{\sim}{r}-\underset{\sim}{p}\right)\times\underset{\sim}{q}=\underset{\sim}{0}$ $\underset{\sim}{q}$ and $\underset{\sim}{r}-\underset{\sim}{p}$ are non-zero $\Rightarrow \quad \underset{\sim}{r}-\underset{\sim}{p}\parallel\underset{\sim}{q}$ $\underset{\sim}{r}-\underset{\sim}{p}=\lambda\underset{\sim}{q},\ \lambda\in\mathbb{R}$ $\underset{\sim}{r}=\underset{\sim}{p}+\lambda\underset{\sim}{q}$ Geometrically, the set of all possible positions of R is the line that passes through point P and is parallel to the vector $\underset{\sim}{q}$.	You **cannot** say the line passes through vector $\underset{\sim}{p}$ (must say point P). [Also, we want the **full line**, so "the set … is on the line…" is wrong.]

(ii) Given instead that $\mathbf{r}=\begin{pmatrix}x\\y\\z\end{pmatrix}$, $\mathbf{p}=\begin{pmatrix}-1\\2\\4\end{pmatrix}$, $\mathbf{q}=\begin{pmatrix}3\\-5\\2\end{pmatrix}$ and that $(\mathbf{r}-\mathbf{p}).\mathbf{q}=0$, find the relationship between x, y and z. Describe the set of all possible positions of the point R in this case. [4]

$\left(\underset{\sim}{r}-\underset{\sim}{p}\right).\underset{\sim}{q}=0$ $\underset{\sim}{r}.\underset{\sim}{q}=\underset{\sim}{p}.\underset{\sim}{q}$ $\begin{pmatrix}x\\y\\z\end{pmatrix}\bullet\begin{pmatrix}3\\-5\\2\end{pmatrix}=\begin{pmatrix}-1\\2\\4\end{pmatrix}\bullet\begin{pmatrix}3\\-5\\2\end{pmatrix}$ $3x-5y+2z=-5$ The set of possible positions of R is the plane that contains the point P and has a normal parallel to $\underset{\sim}{q}$.	Have to describe a **unique** plane (eg. our solution), giving only the **normal** is insufficient. [Also, we want the **full plane**, so "the set … is in the plane…" is wrong.] You cannot say the plane contains $\begin{pmatrix}-1\\2\\4\end{pmatrix}$ (which is a vector) but must say the plane contains $(-1, 2, 4)$ or P.

Similar to Specimen paper

6 The complex number z satisfies the equation

$$z^2(2+i)-8iz+t=0,$$

where t is a real number. It is given that one root is of the form $k+ki$, where k is real and non-zero. Find t and k, and the other root of the equation. [8]

Solutions	Comments
$k+ki$ is a root $\Rightarrow$ $(k+ki)^2(2+i)-8i(k+ki)+t=0$ $k^2(1+i)^2(2+i)-8ki(1+i)+t=0$ $k^2(-2+4i)-8k(-1+i)+t=0$ $(-2k^2+8k+t)+(4k^2-8k)i=0$ Since $k\in\mathbb{R}$, equate real parts and equating imaginary parts: $\begin{cases} -2k^2+8k+t=0 & -----(1)\\ 4k^2-8k=0 & -----(2)\end{cases}$ (2): $4k(k-2)=0$ and $k\neq 0$ $\Rightarrow$ $\underline{\underline{k=2}}$ (1): $-8+16+t=0$ $\Rightarrow$ $\underline{\underline{t=-8}}$ Hence, $z^2(2+i)-8iz-8=0$ $z=\dfrac{8i\pm\sqrt{(-8i)^2-4(2+i)(-8)}}{2(2+i)}$ $z=2+2i$ or $z=-0.4+1.2i$ $\underline{\underline{\text{The other root is } -0.4+1.2i}}$. **Alternative solution:** $z^2(2+i)-8iz-8=0$ $z^2-\dfrac{8i}{2+i}z-\dfrac{8}{2+i}=0$ $z^2-\left(\dfrac{8}{5}+\dfrac{16}{5}i\right)z+\left(-\dfrac{16}{5}+\dfrac{8}{5}i\right)=0$ $2+2i$ is a root. Let the other root be w. $(z-2-2i)(z-w)=0$ Coefficient of z: $(-2-2i)-w=-\dfrac{8}{5}-\dfrac{16}{5}i$ $\therefore$ $w=-0.4+1.2i$ $\underline{\underline{\text{The other root is } -0.4+1.2i}}$.	$k+ki$ is **not** a root! (Because coefficients of equation are not all real.) Use GC where necessary, as this question doesn't disallow that. Again, use GC where necessary. $z^2(2+i)-8iz-8=(z-2-2i)(z-w)$ **is wrong**, but instead it should be $(z-2-2i)(az-w)$, where $a=2+i$ But else, rewrite $z^2(2+i)-8iz-8=0$ as $z^2-\dfrac{8i}{2+i}z-\dfrac{8}{2+i}=0$ first, as in our solution, so that coefficient of $z^2=1$

Similar to 2016/I/Q7

7 Do not use a calculator in answering this question.

It is given that $\mathrm{f}(x)=2-\sin 4x$.

(i) Find $\int \mathrm{f}(x)\,\mathrm{d}x$. [1]

Solutions	Comments
$\int \mathrm{f}(x)\,\mathrm{d}x=\int 2-\sin 4x\,\mathrm{d}x$ $=2x+\frac{1}{4}\cos 4x+C$	Don't forget the arbitrary constant C.

(ii) Find the exact value, in terms of π, of $\int_0^{\frac{1}{2}\pi} x\,\mathrm{f}(x)\,\mathrm{d}x$. [4]

Solutions	Comments
$\int_0^{\frac{1}{2}\pi} x\,\mathrm{f}(x)\,\mathrm{d}x=\left[\left(2x+\frac{1}{4}\cos 4x\right)x\right]_0^{\frac{1}{2}\pi}-\int_0^{\frac{1}{2}\pi} 2x+\frac{1}{4}\cos 4x\,\mathrm{d}x$ $=\left(\pi+\frac{1}{4}\right)\frac{\pi}{2}-\left[x^2+\frac{1}{16}\sin 4x\right]_0^{\frac{1}{2}\pi}$ $=\frac{\pi^2}{2}+\frac{\pi}{8}-\frac{\pi^2}{4}$ $=\frac{\pi^2}{4}+\frac{\pi}{8}$	<u>Integration by parts</u> Between x and f(x), you should choose to integrate f(x) as suggested from **(i)**. Integrating x will only make the expression more complicated.

(iii) Find the exact value, in terms of π, of $\int_0^{\frac{1}{2}\pi} (\mathrm{f}(x))^2\,\mathrm{d}x$. [4]

Solutions	Comments
$\int_0^{\frac{1}{2}\pi} (\mathrm{f}(x))^2\,\mathrm{d}x=\int_0^{\frac{1}{2}\pi} (2-\sin 4x)^2\,\mathrm{d}x$ $=\int_0^{\frac{1}{2}\pi} 4-4\sin 4x+\sin^2 4x\,\mathrm{d}x$ $=\int_0^{\frac{1}{2}\pi} 4-4\sin 4x+\frac{1-\cos 8x}{2}\,\mathrm{d}x$ $=\int_0^{\frac{1}{2}\pi} \frac{9}{2}-4\sin 4x-\frac{1}{2}\cos 8x\,\mathrm{d}x$ $=\left[\frac{9}{2}x+\cos 4x-\frac{1}{16}\sin 8x\right]_0^{\frac{1}{2}\pi}$ $=\left(\frac{9\pi}{4}+1-0\right)-(0+1-0)$ $=\frac{9\pi}{4}$	All steps to be shown clearly since the use of calculator is not allowed.

Similar to 2017/I/Q7

8 (a) The 1st term of an arithmetic series is 4 and the 5th term is 10.

(i) Find the 30th term of this series. [2]

Solutions	Comments
Let u_n be the nth term and let d be the common difference. Given $u_5 = 10$, we have $4+4d=10 \Rightarrow d=\dfrac{3}{2}$. Hence, $u_{30} = 4+29\left(\dfrac{3}{2}\right)=47.5$.	

(ii) Find the sum of the 21st term to the 50th term inclusive of this series. [3]

Solutions	Comments
$u_{21}+u_{22}+...+u_{50} = \dfrac{30}{2}\left(u_{21}+u_{50}\right)$ $= 15\left(4+20\left(\dfrac{3}{2}\right)+4+49\left(\dfrac{3}{2}\right)\right)$ $=1672.5$ **Alternative solution** Let S_n be the sum of the first n terms. $u_{21}+u_{22}+...+u_{50} = S_{50}-S_{20}$ $=\dfrac{50}{2}\left(2(4)+49\left(\dfrac{3}{2}\right)\right)-\dfrac{20}{2}\left(2(4)+19\left(\dfrac{3}{2}\right)\right)$ $=1672.5$	$S_{50}-S_{20}$ $=\left(u_1+u_2+u_3+\ldots+u_{50}\right)$ $-\left(u_1+u_2+u_3+\ldots+u_{20}\right)$ $=u_{21}+u_{22}+...+u_{50}$

(b) The 1st term of a geometric series is 4 and the 5th term is 1.6384 where the common ratio is positive.

(i) Find the sum to infinity of this series. [2]

Solutions	Comments
Let r be the common ratio. Given $u_5 = 1.6384$, we have $4r^4 = 1.6384 \Rightarrow r = \sqrt[4]{\dfrac{1.6384}{4}} = 0.8$ since $r>0$. Hence, sum to infinity $= \dfrac{4}{1-0.8} = 20$.	

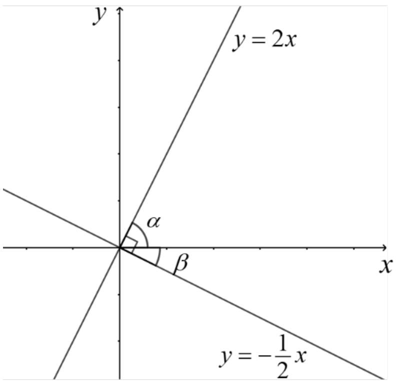
(ii) Given that the sum of the first n terms is greater than 19.6, show that

$$0.8^n < 0.02.$$

Hence find the smallest possible value of n. [5]

Solutions	Comments
$\frac{4(1-0.8^n)}{1-0.8} > 19.6$ $1-0.8^n > \frac{19.6\times 0.2}{4}$ $1-0.8^n > 0.98$ $1-0.98 > 0.8^n$ $0.8^n < 0.02$ (shown) $n\ln 0.8 < \ln 0.02$ $n > \frac{\ln 0.02}{\ln 0.8} = 17.531$ (to 5s.f.) Hence, smallest possible value of n is 18.	Remember to change the inequality sign when dividing by $\ln 0.8$ (which is negative)

9 (i) By considering the gradients of two lines, explain why $\tan^{-1}(2) - \tan^{-1}\left(-\frac{1}{2}\right) = \frac{1}{2}\pi$. [1]

Solutions	Comments
Gradient of line, $m = \tan\theta$ $\theta = \tan^{-1} m$ Hence, gradient of first line, $m_1 = 2$ gradient of second line, $m_2 = -\frac{1}{2}$ Since $m_1 \times m_2 = 2\times\left(-\frac{1}{2}\right) = -1$, both lines are perpendicular to each other. Consider the graphs of $y = 2x$ and $y = -\frac{1}{2}x$  From the diagram, $\alpha = \tan^{-1} 2$ and $\beta = -\tan^{-1}\left(-\frac{1}{2}\right)$. Hence, $\tan^{-1}(2) - \tan^{-1}\left(-\frac{1}{2}\right) = \alpha + \beta = \frac{1}{2}\pi$.	Recall from trigonometry: $\tan\theta = \frac{\text{opposite side}}{\text{adjacent side}}$ $= \frac{\Delta y}{\Delta x}$ $=$ gradient of line, m Note that $\tan^{-1}\left(-\frac{1}{2}\right) < 0$. The gradient of a straight line graph is given by $\tan\theta$ where θ, $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$, is the angle which the line makes with the positive x-direction.

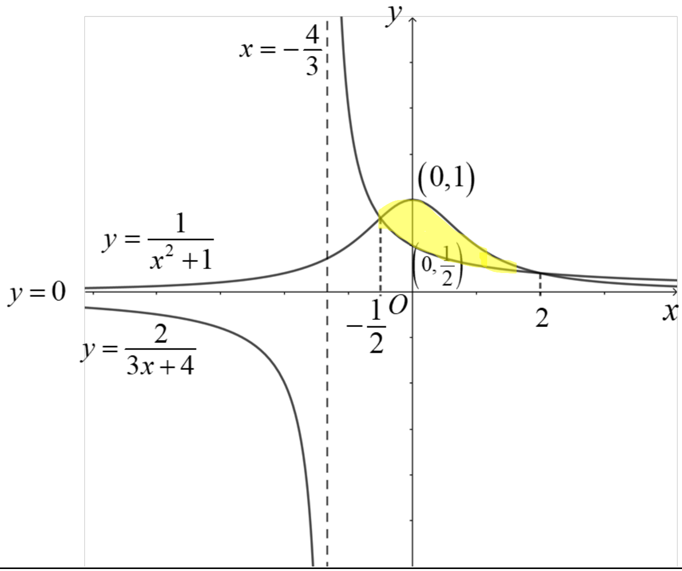
The curves C_1 and C_2 have equations $y=\dfrac{1}{x^2+1}$ and $y=\dfrac{k}{3x+4}$ respectively, where k is a constant and $k>0$.

(ii) Find the set of values of k such that C_1 and C_2 intersect. [3]

Solutions	Comments
$\dfrac{1}{x^2+1}=\dfrac{k}{3x+4}$ $kx^2+k=3x+4$ $kx^2-3x+k-4=0$ For C_1 and C_2 to intersect, the discriminant ≥ 0 and $k\neq 0$. $9-4k(k-4)\geq 0$ $4k^2-16k-9\leq 0$ $(2k+1)(2k-9)\leq 0$ $-\dfrac{1}{2}\leq k\leq\dfrac{9}{2}$ It is also given that $k>0$. Hence, the required set of values of k given by $\left\{k\in\mathbb{R}:0<k\leq\dfrac{9}{2}\right\}$.	Take note of the condition given in the question that $k>0$

It is now given that $k=2$.

(iii) Sketch C_1 and C_2 on the same graph, giving the coordinates of any points where C_1 or C_2 cross the axes and the equations of any asymptotes. [3]

Solutions	Comments
	Must state: (1) coordinates of both y intercepts (2) State the equation of the horizontal & vertical asymptotes.

(iv) Find the exact area of the region bounded by C_1 and C_2, simplifying your answer.[5]

Solutions	Comments
Area $= \int_{-\frac{1}{2}}^{2} \frac{1}{x^2+1} - \frac{2}{3x+4} \, dx$ $= \left[\tan^{-1} x - \frac{2}{3}\ln(3x+4) \right]_{-\frac{1}{2}}^{2} \quad \left(\because 3x+4>0 \text{ for } -\frac{1}{2} \le x \le 2 \right)$ $= \tan^{-1}(2) - \frac{2}{3}\ln 10 - \tan^{-1}\left(-\frac{1}{2}\right) + \frac{2}{3}\ln\frac{5}{2}$ $= \tan^{-1}(2) - \tan^{-1}\left(-\frac{1}{2}\right) + \frac{2}{3}\ln\frac{1}{4}$ $= \frac{\pi}{2} - \frac{2}{3}\ln 4$ (from part (i)) $= \frac{\pi}{2} - \frac{4}{3}\ln 2$	Need to simplify your answer which is to combine the logarithmic terms. Many did not make the connection between this part and the simplification of $\tan^{-1}(2) - \tan^{-1}\left(-\frac{1}{2}\right) = \frac{\pi}{2}$

10 Scientists are investigating the effect of disease on the number of sheep on a small island. They discover that every year the death rate of the sheep is greater than the birth rate of the sheep. The difference every year between the death rate and the birth rate for the population of sheep on the island is 3%. The number of sheep on the island is P at a time t years after the scientists begin observations.

(i) Write down a differential equation relating P and t. [2]

Solutions	Comments
$\frac{dP}{dt} = -0.03P$	*Explanation*: $\frac{dP}{dt} = \text{Rate of birth} - \text{Rate of death}$ (assuming no migration) $\frac{dP}{dt} = -0.03P$ $(\because \text{death rate} > \text{birth rate})$

(ii) Solve this differential equation to find an expression for P in terms of t. Explain what happens to the number of sheep if this situation continues over many years. [4]

Solutions	Comments
$\frac{1}{P}\frac{dP}{dt} = -0.03$ $\int \frac{1}{P} \, dP = \int -0.03 \, dt$ $\ln P = -0.03t + C$ $(\because P>0)$ $P = e^{-0.03t+C}$ $= Be^{-0.03t}$, $B = e^C$ As $t \to \infty$, $e^{-0.03t} \to 0$ and therefore, $P \to 0$. As the situation continues over many years, the number of sheep will (decrease and) tend to 0.	Good practise to use contextual clues ($P > 0$) to remove modulus when taking ln. Alternatively, we can use working similar to iv. Since no boundary conditions were given in the question, general solution is expected Must state the conclusion within the context of sheep.

The scientists import sheep at a constant uniform rate of n sheep per year. (The difference every year between the death rate and the birth rate remains at 3%.)

(iii) Write down a differential equation to model the new situation. [2]

Solutions	Comments
$\frac{\mathrm{d}P}{\mathrm{d}t} = -0.03P + n$	New rate = initial rate + import rate

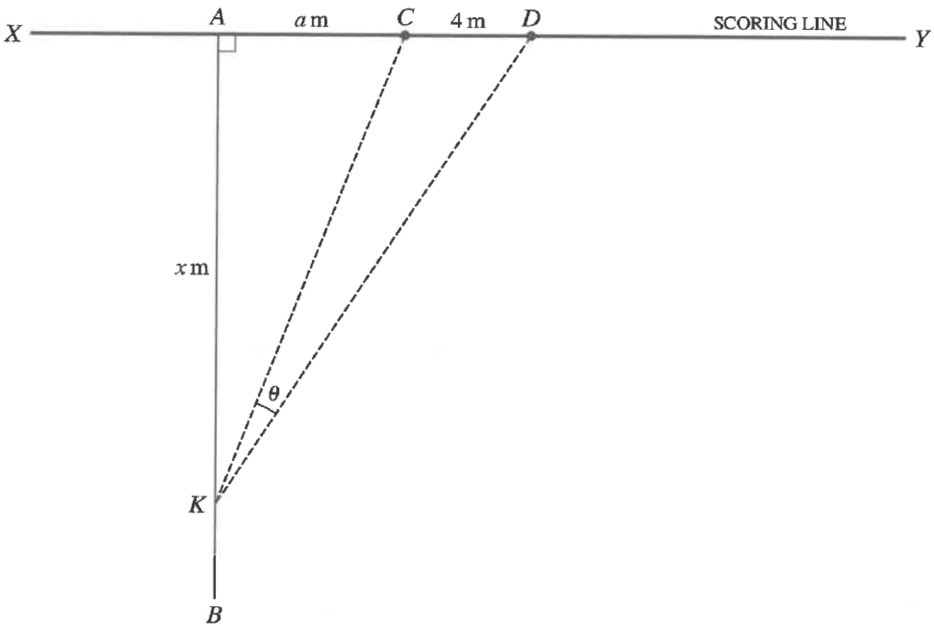
(iv) Solve the differential equation to find an expression for P in terms of t and n. [4]

Solutions	Comments
$\frac{1}{n-0.03P}\frac{\mathrm{d}P}{\mathrm{d}t} = 1$ $\int \frac{1}{n-0.03P}\,\mathrm{d}P = \int 1\,\mathrm{d}t$ $\frac{1}{-0.03}\ln\lvert n-0.03P\rvert = t + C$ $\lvert n-0.03P\rvert = \mathrm{e}^{-0.03t-0.03C}$ $= B\mathrm{e}^{-0.03t}$, $\quad B = e^{-0.03C}$ $n - 0.03P = A\mathrm{e}^{-0.03t}$ $P = \frac{100}{3}\left(n - A\mathrm{e}^{-0.03t}\right)$	Note that n is a constant. Replace B with A after removing modulus where $A = B$ if $n - 0.03P > 0$ **or** $A = -B$ if $n - 0.03P < 0$

(v) Given that the number of sheep settles down to 500 after many years, find n. [2]

Solutions	Comments
As $t \to \infty$, $\mathrm{e}^{-0.03t} \to 0$ and therefore, $P \to \frac{100n}{3}$. $\therefore \frac{100n}{3} = 500$ $n = 15$	

11 In sport science, studies are made to optimise performance in all aspects of sport from fitness to technique.



In a game, a player scores 3 points by carrying the ball over the scoring line, shown in the diagram as XY. When a player has scored these 3 points, an extra point is scored if the ball is kicked between two fixed vertical posts at C and D. The kick can be taken from any point on the line AB, where A is the point at which the player crossed the scoring line and AB is perpendicular to XY.
The distance CD is 4 m; XC is equal to DY; the point A is a distance a m from C and A lies between X and C. The kick is taken from the point K, where AK is x m. The angle CKD is θ (see diagram).

(i) y expressing θ as the difference of two angles, or otherwise, show that

$$\tan\theta = \frac{4x}{x^2+4a+a^2}.$$ [3]

Solutions	Comments
$\begin{aligned} \tan\theta &= \tan\left(A\hat{K}D - A\hat{K}C\right) \\ &= \frac{\tan A\hat{K}D - \tan A\hat{K}C}{1+\tan A\hat{K}D \tan A\hat{K}C} \\ &= \frac{\frac{a+4}{x}-\frac{a}{x}}{1+\left(\frac{a+4}{x}\right)\left(\frac{a}{x}\right)} \\ &= \frac{\frac{4}{x}}{1+\frac{a(a+4)}{x^2}} \\ &= \frac{4x}{x^2+a(a+4)} \\ &= \frac{4x}{x^2+4a+a^2} \quad \text{(shown)} \end{aligned}$	You will need to use the formula from MF26: $\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$

(ii) Find, in terms of a, the value of x which maximises $\tan\theta$, simplifying your answer. Find also the corresponding value of $\tan\theta$. (You need not show that your answer gives a maximum.) [3]

<table>
<tr><th>Solutions</th><th>Comments</th></tr>
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$$\frac{\mathrm{d}(\tan\theta)}{\mathrm{d}x}=\frac{4\left(x^2+4a+a^2\right)-4x(2x)}{\left(x^2+4a+a^2\right)^2}$$

For stationary value of $\tan\theta$, $\dfrac{\mathrm{d}(\tan\theta)}{\mathrm{d}x}=0$.

$$4\left(x^2+4a+a^2\right)-4x(2x)=0$$
$$x^2+4a+a^2-2x^2=0$$
$$-x^2+4a+a^2=0$$
$$x=\sqrt{a^2+4a}\quad(\because x>0)$$

$$\tan\theta=\frac{4\sqrt{a^2+4a}}{a(a+4)+4a+a^2}=\frac{4\sqrt{a^2+4a}}{2\left(a^2+4a\right)}=\frac{2}{\sqrt{a^2+4a}}$$

</td><td>You would need to apply your quotient rule here.
More likely to make algebraic error for those who differentiated by first making θ the subject of the expression. Since the question asks for the maximum value of tan (, there is no need to make (the subject.

Remember to continue to find the corresponding value of $\tan\theta$</td></tr>
</table>

The point corresponding to the value of x found in part **(ii)** is called the optimal point. The corresponding value of θ is called the optimal angle.

(iii) Explain why a player may decide not to take the kick from the optimal point. [1]

Solutions	Comments
Players may be trained to only take kicks from certain fixed points on the field. OR A player may choose to take the kick from a distance closer to the scoring line as it requires less force to score.	

(iv) Show that, when θ is the optimal angle, $\tan KDA=\sqrt{\dfrac{a}{4+a}}$. Find the approximate value of angle KDA when a is much greater than 4. [3]

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When θ is the optimal angle,

$$\tan K\hat{D}A=\frac{x}{a+4}=\frac{\sqrt{a(a+4)}}{a+4}=\sqrt{\frac{a}{4+a}}\quad\text{(shown).}$$

When a is much greater than 4, $\dfrac{a}{4+a}=1-\dfrac{4}{4+a}\approx1-0=1$.

</td><td>You can substitute the value of x into an expression from part (i), there is no need to work all over again.</td></tr>
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$\tan K\hat{D}A = \sqrt{\dfrac{a}{4+a}} \approx 1$ $K\hat{D}A \approx \dfrac{\pi}{4}$	

(v) It is given that the length of the scoring line XY is 50 m. Find the range in which the optimal angle lies as the location of A varies between X and C. [2]

Solutions	Comments
$XY = 50 \Rightarrow 0 < a < \dfrac{50-4}{2} = 23$ This implies that $\tan\theta = \dfrac{2}{\sqrt{a(a+4)}} > \dfrac{2}{\sqrt{23(27)}} \approx 0.080257.$ $\theta > \tan^{-1} 0.080257 = 0.0801 \quad \text{(to 3 s.f.)}$ Since θ is acute, $0.0801 < \theta < \dfrac{\pi}{2}$.	